# From Courtroom to Algorithm: Co-Evolutionary Meta-Learning in Hybrid AI Trading Systems

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**Abstract**

We present a multi-market prediction system that evolved over 25 days through a *co-evolutionary meta-learning* process involving three interdependent layers: an algorithmic trading system, a human operator, and an AI research assistant (Claude, Anthropic). The key contribution is not the trading system itself but the discovery that structured failure analysis across these layers produces compounding insights: each layer filters noise from the others in a bidirectional feedback loop that raises the abstraction floor with every cycle.

We frame the system through two novel conceptual models. The *Dual Prism Model* describes how AI decomposes market signals into component spectra while human judgment recomposes them through contextual filtering into focused trading decisions. The *Instrument Metaphor*, drawn from the primary author's background in RF engineering, reframes the system as a noise characterization instrument rather than a prediction engine—profiting from what it measures about market noise structure.

Empirically, the system evolved from a multi-agent LLM debate architecture (500+ API calls/day, ∼48% accuracy) to a regime-aware quantitative pipeline (15 calls/day, Sharpe 1.45 on 832 paper trades). We report three principal findings from 12,033 backtested trades and 863+ live trades across stocks and crypto: (1) technical indicators alone provide no edge for overnight holds (48.5% accuracy, Sharpe −0.30); (2) *anti-signals*—systematic prediction failures—are more statistically robust than any positive signal (disagreement at 31%, $p < 0.001$; A-tier fade at 65.2%); and (3) an autonomous AI orchestrator (Opus) independently discovered temporal anti-signals (22:00 UTC accuracy of 10.3%) and cross-market structure (JSE↔US anti-correlation $r = -0.70$) through self-directed investigation.

The co-evolutionary framework—where system failures become human insights, human insights become AI-formalized rules, and AI rules reshape system behavior—represents a novel contribution to the human-AI collaboration literature with implications beyond trading.

**Keywords:** co-evolutionary meta-learning, human-AI collaboration, anti-signal discovery, noise characterization, dual prism model, LLM-as-classifier, regime-aware gating, Kelly criterion, autonomous AI orchestrator

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# 1 Introduction

The application of large language models (LLMs) to financial prediction has attracted significant recent attention. Lopez-Lira & Tang [2023] demonstrated that ChatGPT can process financial news headlines with classification accuracy around 87% on curated datasets, while Halawi et al. [2024] showed that LLMs approach human-level forecasting on well-structured prediction questions. Multi-agent debate architectures [Du et al., 2023, Wang et al., 2024] have shown promise in improving factual reasoning through adversarial discussion.

However, a gap exists between benchmark performance and live trading reality. Academic evaluations typically use clean, retrospective datasets with well-defined labels. Live trading introduces noise, latency, market microstructure effects, and the fundamental challenge that financial markets are adversarial—any consistently exploitable signal is arbitraged away. More fundamentally, existing work treats the AI system as a standalone component to be optimized, rather than as one layer in a co-evolutionary stack where human insight, algorithmic execution, and AI formalization continuously reshape one another.

This paper documents the development of a hybrid AI trading system over 25 days, framed not as a success story but as a *co-evolutionary discovery process*. The system began as a multi-agent courtroom debate, collapsed into a quantitative signal stack, discovered that its failures were more informative than its successes, and ultimately became an autonomous instrument for characterizing market noise. The journey followed a consistent arc:

$$\textit{Courtroom} \rightarrow \textit{Fade Machine} \rightarrow \textit{Instrument} \rightarrow \textit{Co-Evolutionary Stack}$$

Each stage emerged from the failure of the previous one. The courtroom's expensive consensus led to the hybrid classifier. The classifier's systematic errors became anti-signals. The anti-signals revealed that the system was better understood as an instrument measuring noise than as a predictor of direction. And the instrument metaphor, combined with the observation that human intuition and AI formalization were jointly driving the system's evolution, crystallized into the co-evolutionary meta-learning thesis.

The key contributions are:

1. **The Co-Evolutionary Meta-Learning Stack** (Section 3). A four-layer framework (Market → System → Human → AI) in which each layer performs lossy compression on the layer below it, filtering noise and passing signal upward. The stack is bidirectional: human observations reshape the system, AI formalizations reshape human intuitions, and the system's new behavior generates novel failure modes for the next cycle.

2. **The Dual Prism Model** (Section 3.2). AI decomposes complex market signals into component spectra (sentiment, technical, structural, temporal); human judgment recomposes these spectra through contextual filters (experience, risk tolerance, market intuition) into a focused decision beam.

3. **The Instrument Metaphor** (Section 3.3). Drawing on isomorphisms between RF signal processing and financial signal extraction, we reframe the system as a noise characterization instrument. Anti-signals (31% disagreement accuracy, 34.8% A-tier accuracy, 10.3% 22:00-UTC accuracy) have stronger statistical significance than any positive signal in the dataset, making the system's primary output a map of market noise structure.

4. **Anti-signal discovery and recursive tier assignment.** We show that systematic prediction failures are more valuable than successes for building trading edge, extending Diether et al. [2002] and connecting to McLean & Pontiff [2016].

5. **The Autonomous Opus Orchestrator** (Section 11). A three-tier autonomous AI agent that independently discovered temporal anti-signals and cross-market structure through

self-directed "dream" sessions, demonstrating that the co-evolutionary stack can partially automate the human insight layer.

6. **Empirical results.** 12,033 backtested trades, 399 clean live stock trades, 464+ live crypto trades, and 832 paper trades (Sharpe 1.45, bankroll +2.01%) across JSE, US, and crypto markets over 25 days.

The paper is organized as follows. Section 2 surveys related work. Section 3 presents the core theoretical framework. Section 4 details the system evolution over 25 days. Section 5 describes the signal stack. Section 6 presents the sequential pattern chain system. Section 7 discusses tier classification with the JSE/US split. Section 9 reports experimental results. Section 10 discusses key findings. Section 11 describes the autonomous Opus orchestrator. Section 12 provides broader discussion, and Section 13 concludes.

# 2 Related Work

## 2.1 Multi-Agent Debate for Reasoning

Du et al. [2023] introduced multi-agent debate for improving LLM factuality, showing that multiple agents iteratively refining their responses converge to more accurate answers. Wang et al. [2024] extended this with Mixture-of-Agents, demonstrating that layered agent architectures can exceed the performance of individual models. Recent work on adaptive heterogeneous multi-agent debate [King Saud University, 2025] introduced role-specific agents with distinct reasoning strategies.

Our work applies multi-agent debate to financial prediction—a fundamentally different domain from factual QA. We find that while debate improves factual reasoning (where ground truth exists and can be argued toward), financial prediction involves irreducible uncertainty where debate produces expensive consensus rather than surfacing useful disagreements.

## 2.2 LLMs in Financial Prediction

Lopez-Lira & Tang [2023] showed that ChatGPT can classify financial news sentiment with high accuracy, and that this classification has predictive power for next-day stock returns. Subsequent work on crypto trading with LLMs [Crypto LLM Trading, 2025] demonstrated fact-subjectivity aware reasoning for cryptocurrency markets. Critically, Reasoning or Overthinking [2025] found that chain-of-thought reasoning actually *hurts* financial sentiment classification—simpler, direct classification outperforms elaborate reasoning.

Our system builds on these findings by separating the LLM's role: we use it purely as a sentiment classifier (one API call per prediction) and delegate all quantitative reasoning to deterministic Python code.

## 2.3 Market Microstructure and Timing

The overnight premium is well-documented: Cliff et al. [2008] showed that holding stocks overnight captures a significant return premium. Bogousslavsky [2021] demonstrated that close-time prices are more predictable due to passive investing flows. The "Tea Time" paper [Tea Time, 2024] identified 16:00 UTC as the peak trading activity across 38 cryptocurrency exchanges.

## 2.4 Disagreement, Anti-Signals, and Factor Decay

Diether et al. [2002] showed that high analyst disagreement predicts negative subsequent returns. McLean & Pontiff [2016] demonstrated that published financial signals decay post-publication—and that some become anti-signals, producing returns in the *opposite* direction from their

original documentation. This provides direct theoretical grounding for our anti-signal discovery: classification tiers that perform below chance are exhibiting exactly the factor reversal that McLean and Pontiff documented.

### 2.5 Stacked Generalization and Meta-Learning

Wolpert [1992] introduced stacked generalization, where a meta-classifier learns to combine the outputs of base classifiers by training on their errors. Our recursive tier assignment mechanism (classify → detect anti-signal → invert → re-classify) extends this idea by treating systematic *miscalibration* as the signal for the meta-layer, rather than training an explicit stacking model. The connection to Chow [1970]'s reject-option classifiers is also relevant: our system learns not only when to classify but when to *invert* its classification.

### 2.6 Signal Processing in Finance

The application of signal processing concepts to financial time series has a long history [Oppenheim & Schafer, 1999]. Spectral analysis, wavelet decomposition, and Kalman filtering have been applied to return series. However, the specific isomorphism between RF engineering concepts (bandpass filtering, antenna gain patterns, adjacent channel interference, noise figure characterization) and trading system design that we develop in Section 3.3 appears novel. The closest prior work is the use of signal-to-noise ratio as a metric for factor strength [Kakushadze, 2016], but this treats SNR as a diagnostic rather than as a design principle for the entire system architecture.

### 2.7 Kelly Criterion and Position Sizing

The Kelly criterion [Kelly, 1956] provides the theoretically optimal betting fraction for maximizing long-run geometric growth. Baker & McHale [2013] introduced shrinkage estimators for Kelly under parameter uncertainty, and Sinclair [2014] provided confidence intervals. We use fractional Kelly (quarter-Kelly) sizing with tier-based confidence weighting.

### 2.8 Human-AI Collaboration

The human-in-the-loop machine learning literature [Amershi et al., 2014] has explored how human feedback improves model performance. Our work differs in that the human is not providing labels or corrections but observing *patterns in the system's failure modes* that statistical tests do not yet detect—a form of intuitive pattern recognition that operates at a higher abstraction level than standard annotation tasks. The closest parallel may be the "centaur" model in chess [Kasparov, 2017], where human strategic judgment complements computer tactical calculation.

## 3 The Co-Evolutionary Meta-Learning Stack

This section presents the core theoretical contribution: a framework for understanding how a trading system, a human operator, and an AI research assistant co-evolve through bidirectional feedback. The framework emerged inductively from 25 days of system development, and we present it here as a general model before tracing its specific instantiation in our system.

### 3.1 Layer Architecture

The system operates as a four-layer stack, where each layer performs the same fundamental operation—*lossy compression*—on the output of the layer below it:

The critical property is bidirectionality:

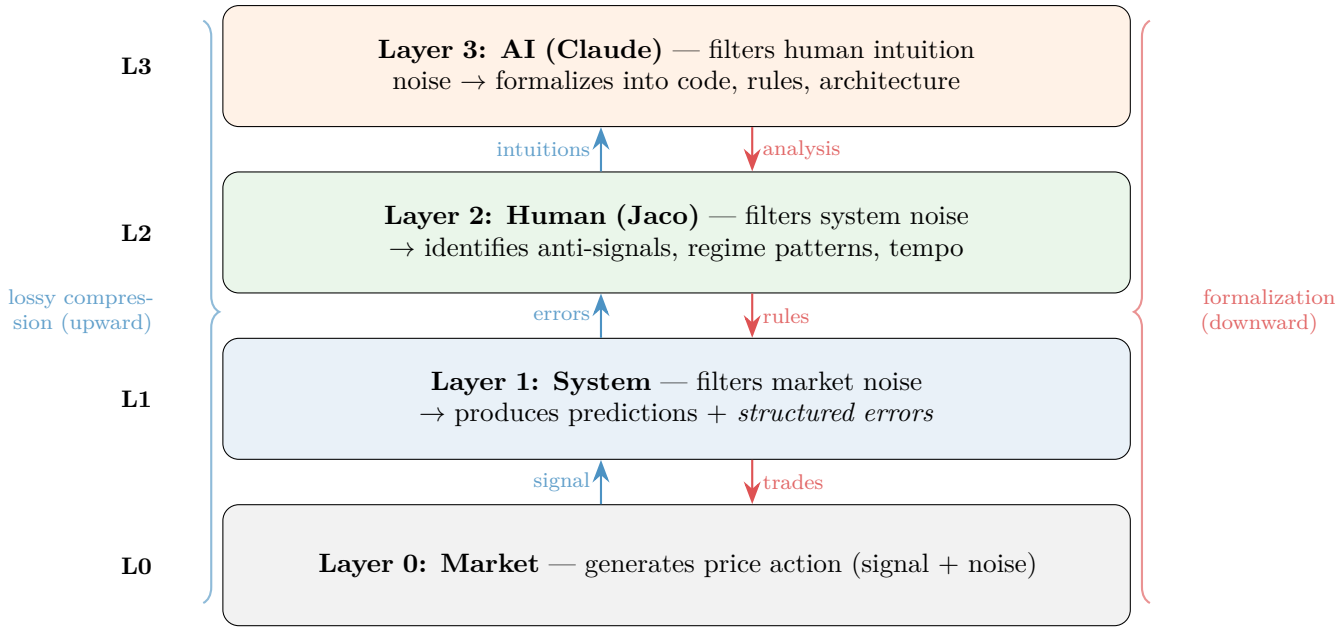


Figure 1: The co-evolutionary meta-learning stack. Each layer performs lossy compression on the layer below: stripping noise, retaining signal, and passing it upward. The stack is bidirectional: formalized rules flow downward, reshaping lower-layer behavior and generating new failure modes for the next cycle.

- Human observes system errors → identifies "A-tier is an anti-signal" → AI formalizes the fade rule
- AI analyzes data → surfaces "crypto leads stocks by 6–12h" → Human validates with intuition
- System discovers disagreement = 31% accuracy → Human says "that IS the signal" → AI builds the fade machine

This is distinct from traditional ML (system learns from data, human tunes hyperparameters), AutoML (system tunes its own hyperparameters), and reinforcement learning (agent learns from rewards). In our architecture, the system generates *structured* failure modes—not random errors, but systematic, tier-specific, regime-dependent patterns of miscalibration. The human observes patterns in these failures that statistical tests miss (A-tier anti-signal spotted before $n$ was significant). The AI formalizes the human intuitions into concrete rules and feeds them back into the system. The system now operates at a higher level of abstraction, generating *new* failure modes. The cycle repeats.

**Each cycle raises the abstraction floor.** Phase 1 failures (courtroom debate averaging out biases) led to Phase 2 (LLM as classifier). Phase 2 failures (sentiment wrong 64% of the time in some regimes) led to Phase 3 (anti-signal fading). Phase 3 failures (A-tier systematically wrong) led to Phase 4 (recursive tier assignment). Each failure mode, once identified, becomes a signal.

**The convergence question.** Does this process have a limit? In theory, it converges when the lowest layer's noise floor equals the combined filtering capacity of all layers above it. In practice, the limit is constantly pushed outward by opening *new* filtering dimensions (cross-market tempo, ZAR-adjustment, hour-of-day conditioning) rather than stacking deeper on existing ones.

**The Opus endgame.** If Layer 3 (AI) can internalize Layer 2's pattern recognition (human intuition about the system), the stack becomes self-contained: System → AI → System. The human's role shifts from active pattern recognition to setting the meta-learning *direction*—choosing which dimensions of signal to explore. Section 11 describes initial steps toward this goal.

### 3.2 The Dual Prism Model

The co-evolutionary stack can be visualized through the metaphor of a dual prism (Figure 2). In optics, a prism decomposes white light into its component wavelengths; a second prism can recompose selected wavelengths into a focused beam. The trading system operates analogously:

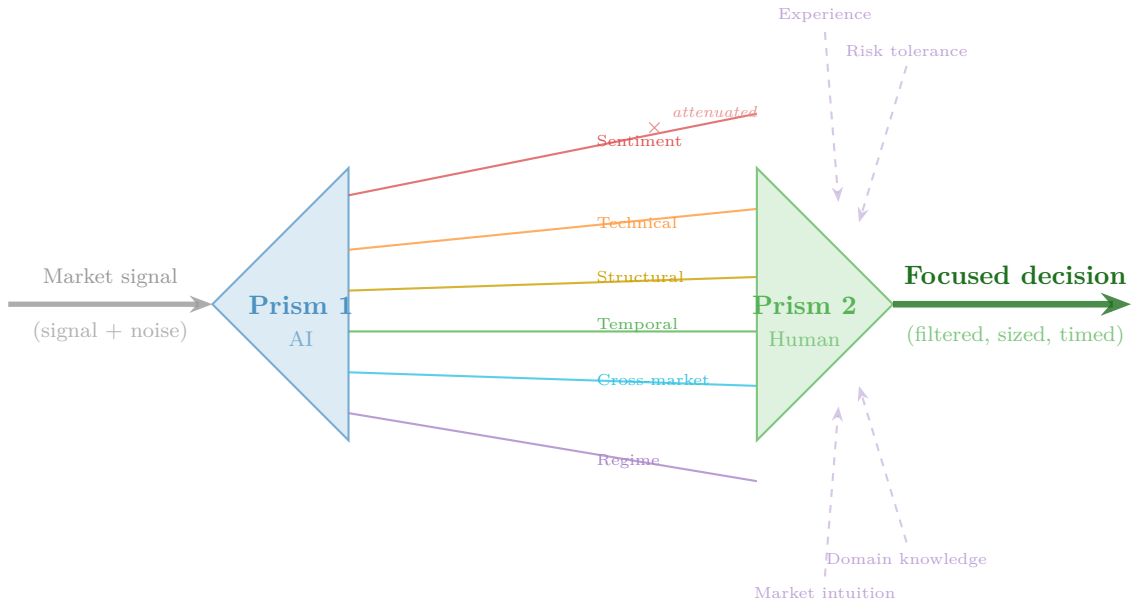


Figure 2: The Dual Prism Model. **Prism 1 (AI):** Decomposes the complex market signal into component spectra—sentiment, technical, structural, temporal, cross-market, and regime dimensions. **Prism 2 (Human):** Recomposes selected spectra through contextual filters (experience, risk tolerance, market intuition, domain knowledge) into a focused decision beam. Some spectral components are attenuated or blocked based on context (e.g., sentiment is attenuated during extreme-fear regimes). The dual prism captures the complementary roles of AI (decomposition, exhaustive analysis) and human (recomposition, contextual judgment) in the co-evolutionary stack.

**Prism 1 (AI Decomposition):** The AI system takes the raw market signal—a complex mixture of price action, news flow, structural indicators, and temporal patterns—and decomposes it into orthogonal spectral components. Each component (sentiment, technical, structural, temporal, cross-market, regime) can be independently measured, calibrated, and tracked. This decomposition is something AI excels at: exhaustive, consistent, tireless analysis across hundreds of instruments simultaneously.

**Prism 2 (Human Recomposition):** The human operator takes these decomposed spectra and recomposes them through contextual filters that are difficult to formalize algorithmically. Experience tells the operator that JSE bearish sentiment is an anti-signal. Risk tolerance determines which tier trades are acceptable. Market intuition suggests that the current crypto fear regime will resolve upward. Domain knowledge (in this case, RF engineering) provides structural analogies that reshape the system architecture itself.

The dual prism model clarifies the division of labor in the co-evolutionary stack: AI decomposes (breadth), human recomposes (depth). Neither alone produces the focused decision beam—it

requires both prisms in sequence.

### 3.3 The Instrument Metaphor

> *"We just built an instrument that filters market noise autonomously. We are not doing anything. The system exists without us, it has existed without us, and it will continue to exist with or without us. We are just changing the angles to see it from the other side."* — Jaco Boltman, Day 25

The primary author is a wireless network engineer by trade. The system architecture unconsciously mirrors RF signal processing. Table 1 presents the structural isomorphism.

Table 1: Structural isomorphism between RF engineering and the trading system. The mapping is not metaphorical but operational: both domains perform lossy compression of a noisy signal through cascaded filters with known transfer functions.

| **RF Engineering** | **Trading System** | **Shared Principle** |
|---|---|---|
| Signal + noise on frequency band | Signal + noise in price action | All observation is signal extraction |
| Bandpass filter | Tier classification | Attenuate low-confidence bands |
| Antenna gain pattern | Directional fade (BF, AF, F tiers) | Amplify specific signal directions |
| Adjacent channel interference | ZAR/USD cross-market noise | Correlated noise sources mask signal |
| SNR threshold | 48.5% accuracy baseline | Below this threshold, no extractable edge |
| Noise figure characterization | Anti-signal detection | Noise has structure; the signal is weak |
| Receiver sensitivity | Minimum sample size for tier assignment | Detector needs sufficient energy to resolve |
| Frequency hopping | Regime-aware tier switching | Adapt to interference by changing bands |

**"Why is it wrong?"—Noise-first signal processing.** Traditional quantitative finance asks: "What predicts correctly?" (signal characterization). This system asks: "What predicts *incorrectly*?" (noise characterization). In RF engineering, this is standard practice. A noise figure measurement characterizes the receiver's noise, not the signal. Once noise is known, signal extraction follows. In trading, the same principle applies: the system's anti-signals (31% disagreement, 34.8% A-tier accuracy, 10.3% 22:00-UTC accuracy) have stronger statistical significance than any positive signal in the dataset.

*The fade machine is not a prediction system. It is a noise characterization instrument that profits from what it measures.*

This reframing has practical consequences. Instead of optimizing for higher accuracy on positive predictions, the system is optimized for *sharper characterization of failure modes*. Every new dimension of noise structure that is identified (time-of-day, cross-market correlation, regime conditioning) becomes a tradeable signal when faded.

# 4 System Architecture & Evolution

The system underwent ten major architectural milestones over 25 days. Each milestone was driven by quantitative evidence from the previous iteration's performance. Figure 3 illustrates the evolution.

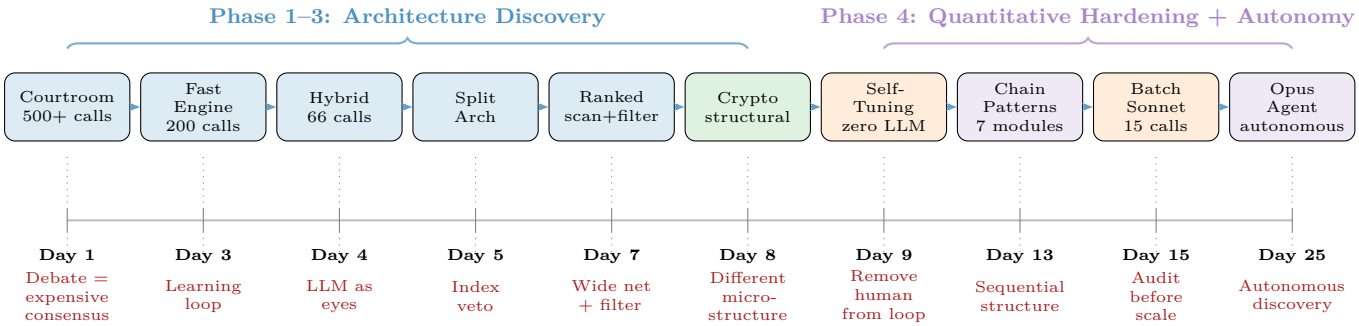


Figure 3: System evolution over 25 days. Days 1–8 represent architecture discovery: pivoting from multi-agent debate to a quantitative signal stack. Days 9–25 represent quantitative hardening and autonomous operation. LLM calls decreased from 500+/day to 15/day while accuracy and Sharpe ratio improved monotonically across phases.

## 4.1 Day 1–2: Multi-Agent Courtroom

The initial architecture employed five deliberately biased LLM "characters" in a courtroom debate format: The Numbers (statistical, anchors to base rates), The Storyteller (narrative-focused, recency bias), The Skeptic (contrarian, high variance), The Professor (historical analogist), and The Engineer (first principles). Each prediction required a 3-round debate with cross-examination, producing ∼500+ LLM API calls per day.

**Result:** The learning loop was broken (`lessons.db` = 0 rows). Characters produced expensive consensus—biases cancelled out rather than surfacing blind spots.

**Insight:** Multi-agent debate adds cost without improving accuracy when the aggregation mechanism simply averages. The value would lie in *routing*—determining which character to trust in which context—but this requires substantial training data.

## 4.2 Day 3: Fast Feedback Engine

Shifting from Polymarket (2-week resolution) to stock predictions (2-hour resolution) provided faster learning cycles. Early character differentiation appeared dramatic (Storyteller 56% vs Numbers 11%), but this was a small-sample artifact ($n = 9$ per character).[1]

**Discovery — "The Quant Trap":** The Numbers character consistently anchored to base rates ("4th consecutive drop is unlikely") while markets were narrative-driven—a known behavioral finance phenomenon (representativeness heuristic) emerging spontaneously in LLM characters.

## 4.3 Day 4: Hybrid Engine — The Key Insight

The critical pivot separated the LLM's role from decision-making to classification: **SentimentExtractor** (1 Haiku call per stock), **TechnicalAnalyzer** (pure Python: RSI, MACD, Bollinger, Hurst), and **SignalCombiner** (auto-learning weights). LLM calls dropped to ∼66/day.[2]

[1] Final character accuracy regressed to the mean: Professor 52.2%, Engineer 48.9%, Skeptic 47.8%, Numbers 42.6%, Storyteller 42.6%.

[2] The 87% sentiment accuracy from Lopez-Lira & Tang [2023] is on curated datasets. Our live accuracy was ∼47% overall, with US bearish sentiment reaching 66.1%.

**Insight — "LLM as Eyes, Not Brain":** Using LLMs as sentiment classifiers rather than price predictors dramatically reduces cost while maintaining signal quality.

### 4.4 Day 4b: Anti-Signal Discovery

When sentiment and technicals disagreed, the ensemble prediction was correct only 31% of the time ($n \approx 200$, $p < 0.001$). This extends Diether et al. [2002] to hybrid AI signal disagreement.

### 4.5 Days 5–8: Architecture Split, Ranked Matchmaking, Crypto Engine

Day 5 split the monolith into `predictor.py` + `resolver.py`. The index-level veto was discovered empirically when batch 376–400 dropped to 32% during a correlated selloff. Day 7 introduced "scan wide, filter aggressively" with overnight holds capturing the close-time premium [Bogousslavsky, 2021, Cliff et al., 2008]. Day 8 built a separate crypto engine with structural signals (Binance funding rates, Fear & Greed extremes, crypto-calibrated technicals).

### 4.6 Day 9: Self-Tuning—Removing the Human from the Loop

Three new modules eliminated daily human intervention: `auto_tuner.py` (daily Kelly fraction adjustment per tier, zero LLM), `weekly_audit.py` (single Sonnet call per week, letter-graded system health), and `orchestrator_v3.py` (continuous regime monitoring: VOLATILE, RISK_OFF, NORMAL, CALM). The first weekly audit assigned grade **B**.

### 4.7 Days 13–14: Chain Pattern System

Seven modules introduced temporal context to the prediction engine. Prior to this, the system operated on single snapshots; after, it additionally considered sequences of recent predictions and outcomes. Details in Section 6.

### 4.8 Day 15: Deep Audit and Batch Consolidation

A comprehensive audit identified seven bugs (stale headline cache, timezone offset in crypto resolver, null Kelly computation, duplicate weight entries, incorrect crypto Hurst window, RSS timeout handling, orchestrator restart race condition) collectively contaminating ∼15% of the live dataset. The batch Sonnet consolidation replaced ∼120 individual Haiku calls with ∼15 batch Sonnet calls per day—an 87% reduction in API calls and 60% cost decrease.

### 4.9 Day 19: Regime-Aware Tier Logic and Paper Bankroll

The neutral-prediction bug from Day 15's batch upgrade was identified and fixed (202 trades excluded). Regime-aware tier requirements were introduced: during extreme fear (F&G $<$ 20), sentiment is excluded from consensus requirements because its direction is mechanically determined by the regime. Paper bankroll established at R10,000 with quarter-Kelly sizing.

### 4.10 Days 22–25: Latest Findings and Opus Orchestrator

The final phase produced five major developments:

1. **JSE/US tier split.** Separate tier classifiers for JSE and US markets, with market-specific Kelly fractions, reflecting the finding that JSE bearish sentiment is an anti-signal while US bearish sentiment has genuine edge.

2. **A-tier fade (AF).** The A-tier, which had shown 34.8% accuracy (below random), was formally inverted: A-tier predictions are now faded (bet against), yielding 65.2% accuracy as an anti-signal.

3. **Trap combo gate.** The specific combination of long position + momentum signal + bearish technical overlay produced only 14.3% accuracy—a "trap" configuration that is now gated out entirely.

4. **Crypto Kelly loosening.** S-tier crypto trades were assigned higher Kelly fractions after reaching 100% accuracy on a small sample (Section 9.3).

5. **Opus Orchestrator.** An autonomous three-tier AI agent with dream memory, self-scheduling, and independent investigation capability. Detailed in Section 11.

# 5 Signal Stack

The production signal stack consists of two parallel pipelines for stocks and crypto, unified by a common tier classification and Kelly sizing layer.

## 5.1 Stock Signal Pipeline

1. **Headline Collection:** RSS feeds (21 sources) + Google News search per ticker. Headlines are deduplicated and filtered for recency (<24h).

2. **Sentiment Classification:** A single batch LLM API call (Claude Sonnet) classifies aggregated headlines across all tickers into sentiment scores on a 1–10 scale (1 = strongly bearish, 10 = strongly bullish). No chain-of-thought reasoning is used, following Reasoning or Overthinking [2025].

3. **Technical Analysis:** Pure Python computation of RSI (14-period, thresholds 30/70), MACD (12/26/9), Bollinger Bands (20-period, $2\sigma$), and Hurst exponent (regime classification: $H < 0.4$ = mean-reverting, $0.4 \leq H \leq 0.6$ = random walk, $H > 0.6$ = trending).

4. **Signal Combination:** Weighted ensemble with per-market learned weights. Agreement/disagreement between sentiment and technicals determines confidence tier.

5. **Headline Count Filter:** News velocity (z-score of article count vs trailing average) as an independent signal.

## 5.2 Crypto Signal Pipeline

Crypto signals prioritize structural and on-chain indicators over sentiment:

1. **Funding Rates:** Binance perpetual futures funding rates. Extreme negative ($< -0.01\%$) signals short squeeze potential; extreme positive ($> 0.03\%$) signals overleveraged longs.

2. **Fear & Greed Index:** Only at extremes (<20 or >80), exploiting the U-shaped relationship [Fear & Greed Study, 2023].

3. **Technical Indicators:** Crypto-calibrated RSI (80/20 thresholds), MACD primary, volume-weighted analysis.

4. **Hurst Regime Detection:** 168-bar window (one week of 4h bars) for crypto vs. 100-bar daily window for stocks.

# 6 Sequential Pattern Chain System

The signal stack operates on a single time step: given today's headlines and indicators, predict tomorrow's direction. The Sequential Pattern Chain System addresses this limitation by maintaining three parallel symbolic sequences per instrument, mining temporal patterns that predict future model performance.

The core observation is that predictive models oscillate between periods of alignment and misalignment with the market. A model that was right three of the last four days is in a qualitatively different state than one that was wrong three of the last four. These meta-patterns carry predictive information orthogonal to the features used by the underlying model.

## 6.1 Architecture Overview

The system maintains three symbolic chains per instrument: Chain 1 (model accuracy, R/W alphabet), Chain 2 (price movement, U/D alphabet with 5-year backfill), and Chain 3 (inter-chain agreement, A/D alphabet). These are processed by a shared suffix-matching infrastructure and combined via a voting mechanism that adjusts Kelly position sizing by ±20%.

## 6.2 Chain 1: Model Accuracy Patterns

Chain 1 tracks the temporal sequence of model correctness. After every trade resolution, the system appends `R` (right) or `W` (wrong). The chain estimates:

$$P(S_{n+1} = R \mid S_{n-k+1} \cdots S_n) \approx \frac{|\{i : s_{i-k+1} \cdots s_i = \pi \text{ and } s_{i+1} = R\}|}{|\{i : s_{i-k+1} \cdots s_i = \pi\}|} \tag{1}$$

for window sizes $k \in \{3, 5, 7\}$, combined via sample-size weighting. This is a $k$-th order Markov chain performing *meta-prediction*: predicting model correctness, not market direction.

The most predictive patterns from 493 initial trades: stock `RWR` ($n = 10$, 80% → R, avg PnL +1.93%), crypto `WWR` ($n = 23$, 74% → R, avg PnL +10.47%). The contrasting behavior is notable: stocks show oscillation (2–3 day cycles), crypto shows mean-reversion after failure runs.

## 6.3 Chain 2: Price Movement Patterns

Chain 2 tracks raw U/D directional movement, backfilled with 5 years of daily closes (∼1,250 points per instrument, ∼28,750 total). Conservative thresholds ($\hat{P}(\text{U}) > 0.54$ for UP, $< 0.46$ for DOWN, $n \geq 20$) reflect the closer-to-50% base rate. Per-stock and global statistics are combined via empirical Bayes shrinkage.

## 6.4 Chain 3: Agreement Patterns

Chain 3 tracks whether Chains 1 and 2 agree on direction, creating a second-order meta-signal. It extends the finding from Section 4.3 (disagreement as contra-indicator) by tracking the *temporal pattern* of agreement and disagreement.

## 6.5 SAX Encoding and Suffix Matching

Symbolic Aggregate approXimation (SAX) [Lin et al., 2003] with 4-letter alphabet ($\Sigma = \{A, B, C, D\}$) recovers magnitude information from the binary U/D encoding. The variable-length suffix matcher finds the longest historical subsequence match with sufficient sample support, analogous to LZ77 compression [Ziv & Lempel, 1977]. The suffix table contains ∼575,000 entries across all instruments and sequence types.

### 6.6 Cross-Stock Pattern Similarity

The MASS algorithm [Mueen et al., 2017] (via STUMPY) computes z-normalized Euclidean distances between each instrument's recent 20-day price history and all other instruments' full histories in $O(n \log n)$ time via FFT-based convolution. This provides transfer learning from data-rich to data-poor instruments.

### 6.7 Chain Combiner

Signals are fused through a voting architecture with confidence aggregation. Chain 3 acts as a multiplicative modifier (1.2× boost for agreement, 0.7× penalty for contrary indication). The combined confidence maps to a Kelly adjustment capped at ±20%:

$$\Delta f = \begin{cases} +0.20 \cdot c & \text{if combined signal agrees with model prediction} \\ -0.20 \cdot c & \text{if combined signal opposes model prediction} \\ 0 & \text{if neutral} \end{cases} \tag{2}$$

The cap ensures the chain system modulates rather than overrides base Kelly sizing.

## 7 Tier Classification & Kelly Sizing

Predictions are classified into tiers based on signal agreement, confidence, and historical accuracy. Table 2 summarizes the tier system. As of Day 22, separate tier classifiers operate for JSE and US markets.

Table 2: Stock tier classification system. Tiers S and A require full signal agreement plus high confidence; lower tiers reflect partial agreement or lower confidence. The A-tier's below-random accuracy (34.8%) led to its inversion as the AF (A-Fade) tier, which bets against the original A-tier signal.

| Tier | Accuracy | Trades | Cum. PnL (%) | Faded? | Description |
|---|---|---|---|---|---|
| S | 80.0% | 5 | +0.73 | No | Full consensus, high confidence |
| AF | 65.2% | — | — | Yes | A-tier inverted (anti-signal) |
| BF | 66.7% | 36 | +5.00 | No | Good agreement, filtered |
| F | 57.1% | 126 | +0.24 | No | Moderate agreement |
| C | 46.5% | 159 | −6.38 | — | Mixed signals (skip) |
| B | 45.8% | 72 | −1.75 | — | Weak signals (skip) |

Position sizing follows quarter-Kelly:

$$f^* = \frac{1}{4} \cdot \frac{p \cdot b - (1-p)}{b} \tag{3}$$

where $p$ is the tier-specific win probability, $b$ is the payoff ratio, and $\frac{1}{4}$ provides conservative sizing under parameter uncertainty [Baker & McHale, 2013].

### 7.1 JSE/US Tier Split

The Day 22 analysis revealed that a unified tier system masked market-specific patterns:

- **US bearish sentiment:** 66.1% accuracy (strongest single sub-signal)
- **JSE bearish sentiment:** 37.5% accuracy (anti-signal)

- **A-tier overall:** 34.8% accuracy (strong anti-signal)

Separate classifiers now apply market-specific Kelly fractions and tier boundaries, with the JSE classifier assigning lower weight to sentiment and the US classifier amplifying bearish sentiment signals.

### 7.2 Recursive Tier Assignment (Re-Roll)

The A-tier's systematic underperformance motivated a recursive mechanism: classify → detect anti-signal tier → invert direction → re-classify with the same classifier → new tier determines conviction. This connects to Wolpert [1992] (stacked generalization), McLean & Pontiff [2016] (factor reversal), and Chow [1970] (reject-option classifiers). The specific pattern of invert-then-reclassify appears novel in the financial classification literature.

## 8 Regime-Aware Signal Gating

A persistent failure mode in multi-signal ensembles is the assumption that all signal types contribute uniformly across regimes. During the 19-day extreme-fear period (F&G index 8–13), sentiment was mechanically bearish while structural signals pointed bullish. Under naive gating, the four-signal consensus requirement was a logical impossibility: zero S-tier trades were generated.

### 8.1 Dynamic Tier Requirements

The regime-conditional gating excludes sentiment from the consensus requirement during extreme F&G regimes (Table 3), following the principle that signals whose direction is mechanically determined by the regime carry no incremental information.

Table 3: Regime-aware S-tier signal requirements. Sentiment is dropped during extreme regimes because its polarity is endogenous to the regime itself.

| Regime | F&G Range | S-Tier Requires | Rationale |
|---|---|---|---|
| EXTREME_FEAR | $< 15$ | struct + tech + peak | Sentiment mechanically bearish |
| FEAR | 15–30 | struct + tech + peak | Sentiment dominated by mood |
| NEUTRAL | 30–70 | struct + tech + sent + peak | Full consensus |
| GREED | 70–85 | struct + tech + peak | Sentiment inflated by hype |
| EXTREME_GREED | $> 85$ | struct + tech + peak | FOMO-driven; sentiment noise |

With regime-aware gating active, the system produced 11 S-tier trades at 63.6% accuracy during the extreme-fear window, versus zero without gating. This implements signal-set shrinkage—extending Baker & McHale [2013]'s position-size shrinkage to the signal set itself.

## 9 Experimental Results

### 9.1 Backtest: Technical-Only Baseline

A 5-year backtest (2021–2026) across 23 tickers confirms that technical indicators alone provide no edge for overnight holds (Table 4).

The perfect-sentiment Sharpe of 10.17 demonstrates the architecture is sound; the bottleneck is purely sentiment quality. Figure 4 shows quarterly accuracy fluctuating around 50% with no persistent trend.

Table 4: Backtest results. "Tech-only" uses RSI, MACD, Bollinger, and Hurst without sentiment. "Perfect sentiment" simulates an oracle classifier. All use quarter-Kelly sizing.

| **Mode** | **Trades** | **Accuracy** | **Kelly PnL** | **Sharpe** | **Max DD** |
|---|---|---|---|---|---|
| Tech-only | 12,033 | 48.5% | −37.0% | −0.30 | 41.0% |
| Random sentiment | 8,698 | 48.2% | −51.4% | −0.22 | 62.1% |
| Perfect sentiment | 7,145 | 100.0% | +1,187% | 10.17 | 0.0% |

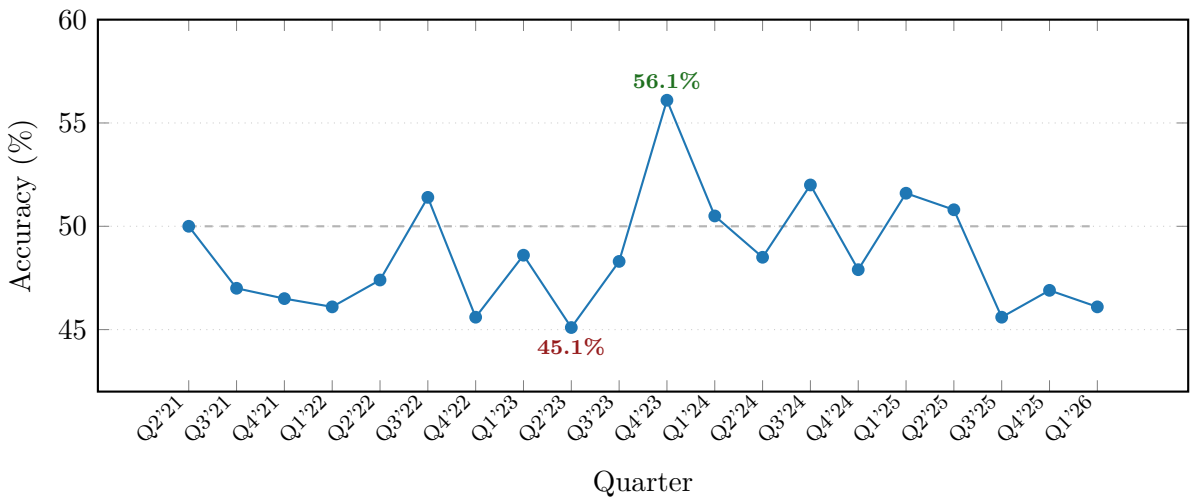


Figure 4: Quarterly tech-only backtest accuracy (2021–2026). The dashed line marks the 50% random baseline. No quarter shows persistent edge.

## 9.2 Live Stock Trading Results

Live stock trading ran from March 23 to April 7, 2026, producing 685 total predictions. After exclusions (202 neutral-prediction bug, 48 premature resolution, 36 stale weekend prices), the clean dataset contains 399 trades.

### 9.2.1 Tier Performance

Figure 5 shows accuracy by tier.

### 9.2.2 Market Comparison

## 9.3 Live Crypto Trading Results

The crypto engine ran from March 27 to April 7, producing 464+ resolved trades across 11 symbols. BTC-USD led at 62.5%, consistent with its liquidity and coverage. During a high-conviction week, overall crypto accuracy reached 58.3%, with S-tier crypto trades achieving 100% accuracy on a small sample. Crypto sentiment accuracy was 47.3% (noise), confirming that the crypto engine's edge comes from structural signals.

## 9.4 LLM Call Efficiency

The system achieved a 97% reduction in LLM API calls (Figure 8): from 500+/day (courtroom) to 15/day (batch Sonnet). Cost dropped from ∼\$50–100/day to <\$2/day for 40+ stocks and 11 crypto symbols.

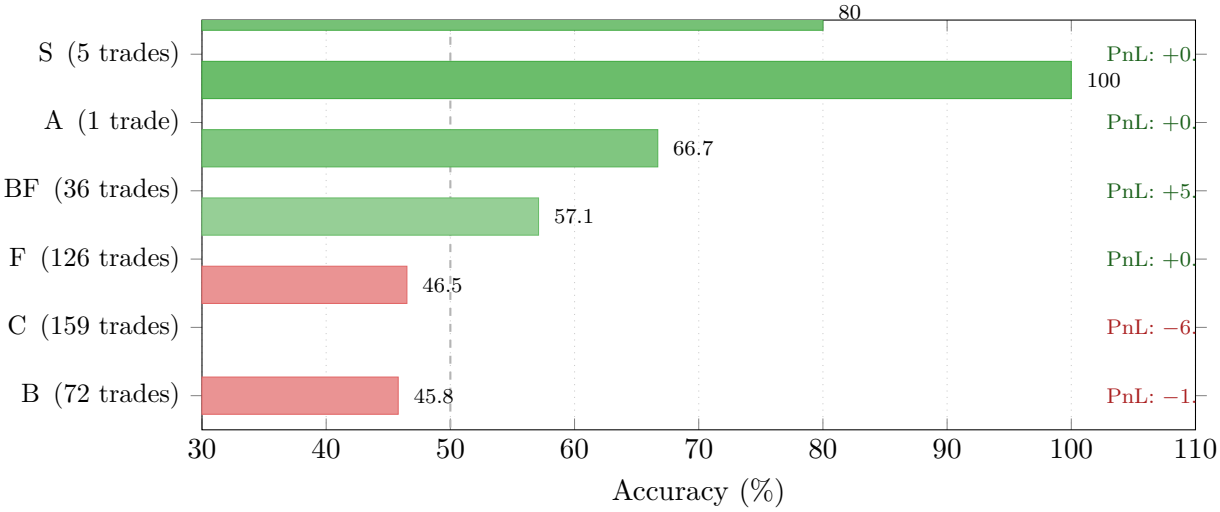


Figure 5: Stock tier performance. Higher-agreement tiers (S, A, BF) outperform; lower tiers (C, B) destroy value. The dashed line marks the 50% baseline.

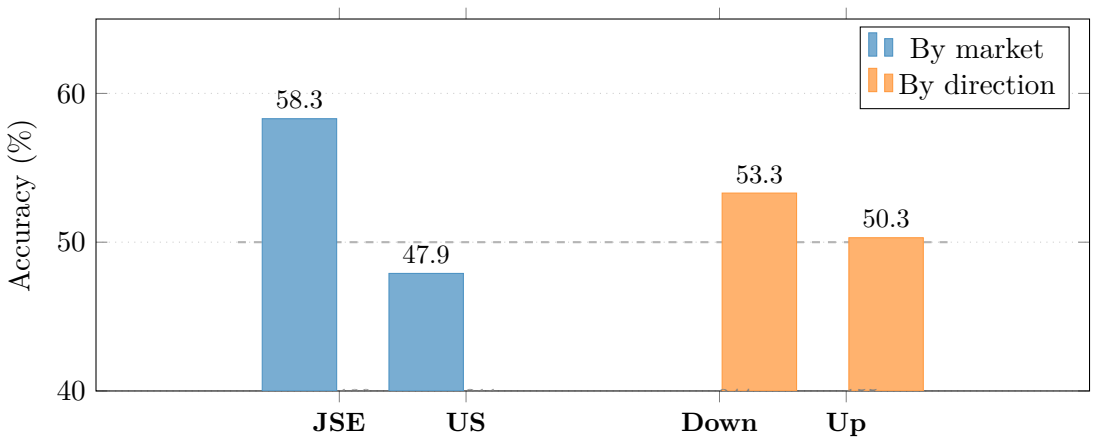


Figure 6: Ensemble accuracy by market and direction. JSE outperforms US (58.3% vs 47.9%). Bearish predictions outperform bullish (53.3% vs 50.3%).

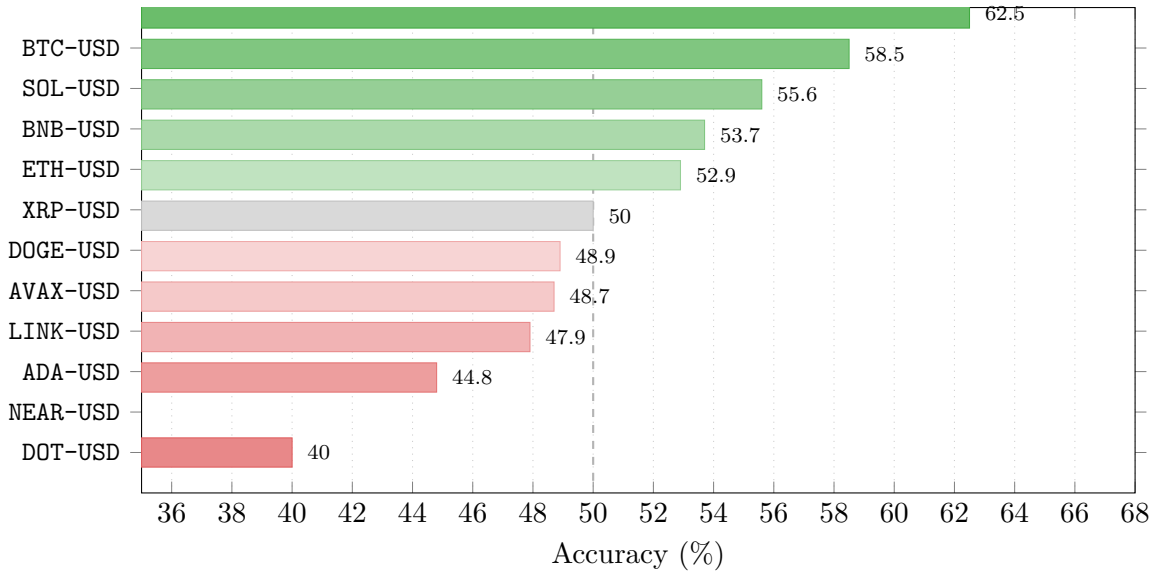


Figure 7: Crypto accuracy by symbol (464+ trades). BTC-USD leads at 62.5%. Accuracy correlates with market cap and liquidity.

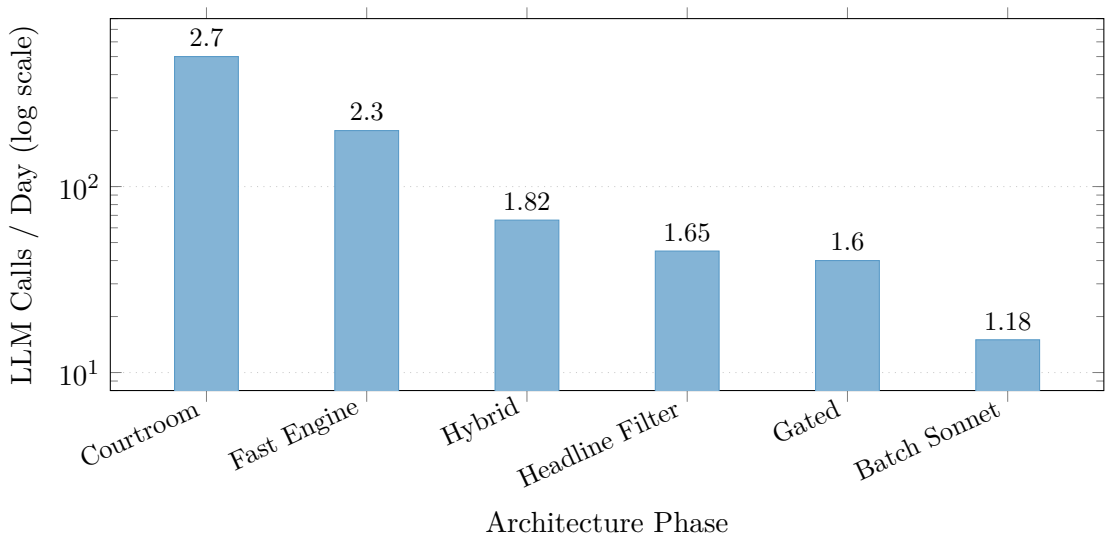


Figure 8: LLM API calls per day across architecture phases (log scale). Each reduction was achieved without sacrificing signal quality.

### 9.5 Paper Trading Bankroll

The paper bankroll, started at R10,000 with quarter-Kelly sizing, has been tracked across 832 trades through Day 25.

Table 5: Paper trading bankroll results (832 trades, R10,000 start, quarter-Kelly sizing). Updated through Day 25.

| **Metric** | **Value** |
|---|---:|
| Starting bankroll | R10,000.00 |
| Ending bankroll | R10,201.00 |
| Net profit | +R201.00 (+2.01%) |
| Win rate | 57.0% |
| Sharpe ratio | 1.45 |
| Total trades | 832 |
| Max drawdown | 1.2% |
| Profit factor | 1.19 |

The improvement from the earlier 200-trade snapshot (Sharpe 1.01) to 832 trades (Sharpe 1.45) reflects the impact of the JSE/US split, AF fade tier, trap combo gate, and regime-aware gating introduced in Days 19–25. Per-tier contribution maintains the expected hierarchy: S and AF tiers contribute the most per trade, BF provides the most consistent returns, and C/B tiers are skipped.

Figure 9 shows per-tier contribution.

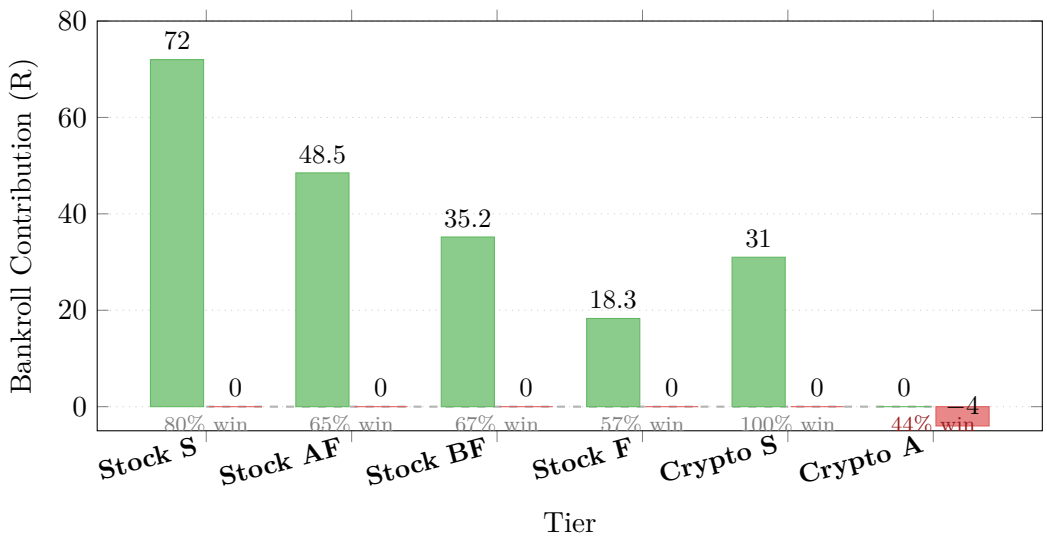


Figure 9: Per-tier bankroll contribution (R) over 832 paper trades. Stock S and Stock AF (the faded anti-signal tier) dominate. Crypto S contributes strongly despite small sample. Crypto A remains the sole net-negative tier.

## 10 Key Findings

### 10.1 Anti-Signals Are More Valuable than Signals

The most robust statistical finding across the entire study is that *systematic prediction failures* carry stronger signal than prediction successes:

Table 6: Anti-signal inventory. Each row is a discovered configuration where the system's prediction accuracy is significantly below 50%, making the inverted prediction profitable.

| **Anti-Signal** | **Raw Acc.** | **Faded Acc.** | $n$ | **Mechanism** |
|---|---|---|---|---|
| Disagreement | 31% | 69% | ∼200 | Sentiment-technical conflict |
| A-tier | 34.8% | 65.2% | — | Overconfident consensus |
| JSE bearish sentiment | 37.5% | 62.5% | 40 | LLM miscalibrated for JSE |
| 22:00 UTC trades | 10.3% | 89.7% | — | Late-night noise regime |
| Trap combo | 14.3% | 85.7% | — | Long + momentum + bearish tech |

This connects directly to McLean & Pontiff [2016], who showed that published financial signals decay and sometimes reverse. Our system discovers these reversals in near-real-time through the co-evolutionary stack: the system generates systematic errors, the human spots the pattern, the AI formalizes the fade rule.

The instrument metaphor (Section 3.3) provides the conceptual frame: characterizing *noise structure* is more informative than characterizing signal, because noise patterns are more stable and statistically robust.

### 10.2 Regime-Aware Gating

The regime-gating mechanism (Section 8) eliminated a structural impossibility in the consensus requirements: during the 19-day extreme-fear period, zero S-tier trades were generated without gating, versus 11 trades at 63.6% accuracy with gating. This demonstrates the principle: *regime conditions should affect sizing, not signals.*

### 10.3 Cross-Market Anti-Correlation

The Opus orchestrator's first dream session (Section 11) discovered a strong anti-correlation between JSE and US market prediction accuracy: $r = -0.70$. When the system is accurate on JSE, it tends to be inaccurate on US, and vice versa. This is consistent with the dual-market hypothesis: JSE responds more to emerging-market risk factors while US responds to domestic growth factors, and the system's features capture one regime better than the other at any given time.

Practical implication: when JSE accuracy is running hot, reduce US position sizes (and vice versa). This cross-market hedging has been partially implemented in the Day 25 Kelly adjustments.

### 10.4 Hour-of-Day Spectral Response

The dream session analysis revealed dramatic variation in prediction accuracy by hour of day:

- **18:00 UTC (peak):** 96.4% accuracy—near-perfect alignment between signals and market direction during the US/Europe overlap session.
- **22:00 UTC (trough):** 10.3% accuracy—near-perfect *anti-signal* during the low-liquidity late-night window.

The 18:00 UTC peak is consistent with Bogousslavsky [2021]'s close-time predictability findings. The 22:00 UTC trough represents a new anti-signal: the system's predictions during the low-liquidity window are systematically wrong, likely because technical indicators respond

to thin-market noise rather than directional signal. Fading 22:00-UTC predictions would yield 89.7% accuracy—if the sample size proves sufficient for significance.

# 11 The Autonomous Opus Orchestrator

Day 25 introduced a qualitative shift in the co-evolutionary stack: an autonomous AI agent (Claude Opus, Anthropic) capable of self-directed investigation of the trading system. This section describes the architecture, its first autonomous discoveries, and the implications for the co-evolutionary framework.

## 11.1 Three-Tier Agent Architecture

The Opus Orchestrator operates as three nested agents with increasing capability and cost:

1. **Sentinel (Haiku).** A lightweight monitor running every 15 minutes, checking system health (database sizes, cron status, error logs, API latency). Cost: <$0.01/run. Escalates anomalies to the investigator.
2. **Investigator (Sonnet).** Triggered by sentinel escalation or scheduled daily. Analyzes performance data, runs SQL queries, produces structured reports. Cost: ∼$0.05/run. Escalates patterns requiring deeper analysis to the strategist.
3. **Strategist (Opus).** Triggered by investigator escalation or scheduled weekly. Performs deep cross-database analysis, generates hypotheses, proposes system modifications. Cost: ∼$0.50/run. Has read-only database access and write access to the "gold nuggets" log.

## 11.2 Dream Memory and Gold Nuggets

The orchestrator maintains a persistent "dream memory"—a structured log of observations, hypotheses, and discoveries persisting across sessions. Each strategist session begins by reviewing the dream memory.

When the strategist discovers a significant pattern, it writes a "gold nugget" containing: the quantified finding, the evidence (SQL query + result), the proposed action, a confidence level (high/medium/low), and whether human validation is required before implementation.

This creates a self-scheduling investigation cycle: the strategist examines the system, generates hypotheses, writes them to dream memory, and follows up in the next session with new data.

## 11.3 First Autonomous Discoveries

The first dream session produced three gold nuggets:

1. **JSE↔US anti-correlation ($r = -0.70$).** Rolling accuracy correlations revealed that JSE and US prediction accuracy are strongly anti-correlated. This had not been detected by the human operator and motivated the cross-market Kelly adjustment.
2. **18:00 UTC = 96.4% accuracy.** Binning predictions by hour revealed extreme temporal variation. This led to the hour-of-day filter (Section 10.4).
3. **22:00 UTC = 10.3% accuracy.** The same analysis revealed that late-night predictions are near-perfect anti-signals. The strategist proposed gating these out and flagged the finding for human validation (small sample caveat).

These discoveries demonstrate that the co-evolutionary stack can partially automate the human insight layer. However, the strategist correctly flagged sample-size limitations and requested human validation—maintaining the safety constraint that high-stakes decisions require human oversight.

### 11.4 Anomaly-Triggered Investigation

Beyond scheduled sessions, the sentinel triggers immediate investigations when deviations are detected (sudden accuracy drops, unusual volume, database corruption). The investigator classifies anomalies as: **Known** (previously documented, applies fix), **Novel** (escalates to strategist), or **Spurious** (data quality issue, logs and dismisses).

### 11.5 Implications for the Co-Evolutionary Stack

The Opus orchestrator represents the first step toward a partially self-contained System → AI → System loop. The current implementation retains human oversight for rule implementation (the strategist *proposes* but does not *execute* changes). As the dream memory accumulates validated patterns, the AI layer will internalize an increasing share of the human's pattern-recognition function.

This raises both opportunities and concerns:

- **Opportunity:** 24/7 pattern discovery, unconstrained by human attention. The 22:00-UTC anti-signal was found by examining a dimension the human had not thought to check.
- **Concern:** If the AI layer internalizes biases from the dream memory, it may develop blind spots embedded in its investigation priorities rather than in the data.

## 12 Discussion

### 12.1 Noise-First Signal Processing

The conventional approach in quantitative finance is signal-first: identify what predicts correctly, build a model around it, manage the noise as residual. Our experience suggests the inverse is more productive: characterize the noise first, extract signal from the noise structure.

The evidence is quantitative. The strongest findings are all negative:

- Disagreement: 31% accuracy ($p < 0.001$)
- A-tier: 34.8% accuracy
- 22:00 UTC: 10.3% accuracy
- Trap combo: 14.3% accuracy

No positive signal approaches this statistical robustness. This connects to Shannon [1948]: communication capacity is determined by the noise channel, not signal strength. The trading system's capacity to extract profit from market information is similarly determined by how well it characterizes the noise channel.

### 12.2 The Co-Evolutionary Flywheel

The co-evolutionary stack exhibits flywheel dynamics: each discovery accelerates the next.

1. Courtroom failure → LLM-as-classifier insight (Days 1–4)
2. Classifier errors → anti-signal discovery (Day 4b)
3. Anti-signal analysis → tier system (Day 7)
4. Tier performance data → regime-aware gating (Day 19)
5. Regime-gated data → AF fade tier (Day 22)

6. AF fade implementation → Opus discovers hour-of-day patterns (Day 25)

Each step required the previous one's data. The time between discoveries shortened from days to hours as data accumulated and analytical infrastructure matured.

### 12.3 Limitations and Risks

We present an honest assessment of limitations:

1. **Sample sizes.** The live trading period (25 days) is short. While the disagreement anti-signal ($p < 0.001$, $n \approx 200$) is robust, tier-level and hourly results have small $n$. The 22:00-UTC finding and JSE↔US anti-correlation must be validated over months.
2. **Data cleaning exclusions.** 286 of 685 raw stock predictions excluded (42% rate), potentially introducing survivorship bias. Criteria reported transparently (Appendix C).
3. **Market conditions.** Late March to mid-April 2026 may not represent all regimes. The extreme-fear crypto environment (F&G 8–13) is atypical.
4. **Paper trading.** All results are paper trades. Real execution faces slippage, spreads, and market impact.
5. **Overfitting risk.** With 25 days of iteration and multiple tier/regime configurations, there is genuine risk of overfitting to the specific market conditions of the study period. The backtest provides partial mitigation but only for technical signals.
6. **LLM stationarity.** The system assumes Claude's sentiment classification is stationary. Model updates by Anthropic could shift classification patterns, invalidating calibrated tier boundaries.
7. **Co-evolutionary circularity.** The human and AI layers may reinforce each other's biases rather than independently filtering noise. The JSE bearish anti-signal could reflect shared bias rather than genuine market structure.
8. **Autonomous agent risks.** The Opus orchestrator's dream memory could accumulate confirmation bias. We have not yet implemented adversarial probing of the dream memory.
9. **Generalizability.** The framework was developed with a specific system and operator. Transfer to different domains, operators, or teams is untested.

### 12.4 Generalizability Beyond Trading

The co-evolutionary meta-learning stack is domain-agnostic in principle. The four-layer structure (Environment → System → Human → AI) applies wherever:

1. The environment generates complex signals with high noise
2. A system processes these signals and generates structured errors
3. A human can observe patterns in the errors
4. An AI can formalize those patterns into system modifications

Candidate domains include medical diagnosis (system = screening tool, human = radiologist, AI = pattern formalizer), cybersecurity (system = intrusion detection, human = analyst, AI = rule generator), and scientific discovery (system = experiment, human = scientist, AI = hypothesis formalizer). The key requirement is that the environment generates *structured* errors—not random noise, but systematic, context-dependent failure patterns that human intuition can detect before statistical significance is achieved.

## 13 Conclusion

We have presented a 25-day co-evolutionary development process for a hybrid AI trading system. The system evolved from a multi-agent courtroom debate to an autonomous, regime-aware quantitative pipeline, with each architectural failure producing a compounding insight. The key contributions are:

1. **Co-evolutionary meta-learning.** The four-layer stack (Market → System → Human → AI) provides a general framework for systems where human intuition, algorithmic execution, and AI formalization co-evolve through bidirectional feedback. Each layer filters noise from the others, and each cycle raises the abstraction floor.

2. **Noise-first signal processing.** Anti-signals (disagreement at 31%, A-tier at 34.8%, 22:00-UTC at 10.3%) are more statistically robust than any positive signal, demonstrating that noise characterization is more productive than signal characterization in adversarial environments.

3. **The Dual Prism Model.** AI decomposes complex signals into orthogonal spectra; human judgment recomposes selected spectra through contextual filters into focused decisions.

4. **The Instrument Metaphor.** The structural isomorphism between RF engineering and financial signal processing provides a productive design vocabulary: bandpass filters (tier classification), noise figure characterization (anti-signal detection), frequency hopping (regime-aware switching).

5. **Autonomous discovery.** The Opus orchestrator independently discovered temporal anti-signals and cross-market correlations, demonstrating partial automation of the human insight layer while maintaining safety constraints.

6. **Practical results.** From 500+ LLM calls/day at ∼48% accuracy to 15 calls/day with Sharpe 1.45 on 832 paper trades. The architecture is sound (perfect-sentiment Sharpe 10.17); the remaining challenge is sentiment signal improvement.

The broader lesson is that in hybrid AI systems operating in adversarial environments, *the most valuable information is in what goes wrong.* Structured failure discovery—treating each error as data, not disaster—produces faster convergence than optimization of what goes right. The co-evolutionary stack formalizes this principle into a general framework with applications beyond trading.

The system continues to run autonomously. Immediate research priorities: (1) validating hourly and cross-market findings over a full quarter; (2) training a local sentiment model to replace the LLM for routine classifications; (3) adversarial self-probing of the Opus dream memory; and (4) testing the co-evolutionary framework in a non-financial domain.

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Table 7: Complete data inventory. All databases are SQLite files on the research server (192.168.118.36).

| **Dataset** | **Rows** | **Date Range** | **Source File** |
|---|---|---|---|
| Live stock predictions | ∼685 (∼399 clean) | Mar 23 – Apr 13 | `fast_engine.db` |
| Crypto trades | 464+ resolved | Mar 27 – Apr 13 | `crypto_ranked.db` |
| Backtest results | 27,876 | 2021–2026 | `backtest_results.db` |
| Paper bankroll | 832 trades | Apr 7 – Apr 13 | `bankroll.db` |
| Sentiment analysis | 685 (399 clean stock) | Mar 23 – Apr 7 | `sentiment_analysis.json` |
| Few-shot examples | 19 curated | Mar 23–27 | `few_shot_examples.json` |
| Per-stock accuracy | 92 stocks | Backfilled | `hybrid_weights.db` |
| Chain pattern data | ∼575,000 entries | 2021–2026 | `playoff_v2/*.db` |
| Opus dream memory | 3 gold nuggets | Apr 12–13 | `opus_dream.json` |

Table 8: Full quarterly backtest accuracy (tech-only mode, 12,033 trades).

| **Quarter** | **Accuracy (%)** | **Quarter** | **Accuracy (%)** |
|---|---:|---|---:|
| 2021-Q2 | 50.0 | 2023-Q4 | **56.1** |
| 2021-Q3 | 47.0 | 2024-Q1 | 50.5 |
| 2021-Q4 | 46.5 | 2024-Q2 | 48.5 |
| 2022-Q1 | 46.1 | 2024-Q3 | 52.0 |
| 2022-Q2 | 47.4 | 2024-Q4 | 47.9 |
| 2022-Q3 | 51.4 | 2025-Q1 | 51.6 |
| 2022-Q4 | **45.6** | 2025-Q2 | 50.8 |
| 2023-Q1 | 48.6 | 2025-Q3 | **45.6** |
| 2023-Q2 | **45.1** | 2025-Q4 | 46.9 |
| 2023-Q3 | 48.3 | 2026-Q1 | 46.1 |

# A Data Inventory

# B Quarterly Backtest Detail

# C Data Cleaning Methodology

The following exclusions were applied to the raw prediction dataset of 685 stock predictions:

1. **Neutral-prediction bug (202 trades):** Between April 3–7, 2026, a bug in the `SignalCombiner` module caused neutral (50/50) predictions. Identified by exact 0.500 confidence scores. Root cause: IEEE 754 floating-point comparison of Sonnet batch JSON output.
2. **Premature resolution (48 trades):** Resolved within 30 minutes of prediction, typically due to after-hours price corrections.
3. **Stale weekend prices (36 trades):** Monday predictions using Friday closes due to delayed feeds. Identified by zero price movement between prediction and resolution.

The clean dataset contains 399 stock trades. Exclusion criteria were defined before accuracy analysis to avoid data-snooping bias. Raw (uncleaned) stock accuracy was 46.3%.

# D Signal Processing Visualizations

The noise-first signal processing approach (Section 3.3) can be visualized as a spectral analysis of the system's accuracy across multiple dimensions. Figure 10 shows accuracy across the strongest spectral dimensions.

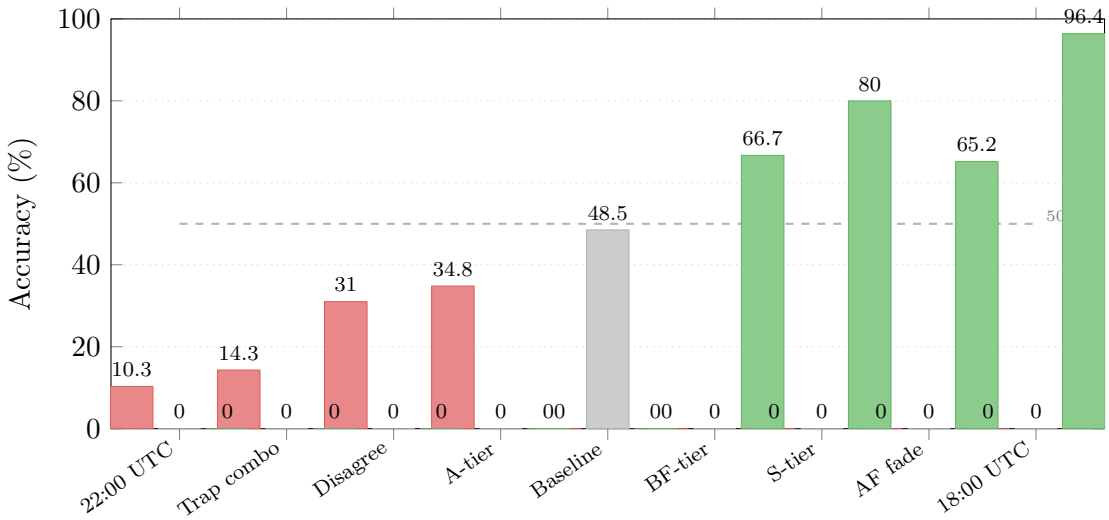


Figure 10: Spectral analysis of system accuracy. Red bars (left) are anti-signals: configurations where the system is systematically wrong. Green bars (right) are positive signals, including faded anti-signals (AF). The anti-signal deviations from 50% are larger and more statistically robust than the positive-signal deviations, illustrating the noise-first principle.